

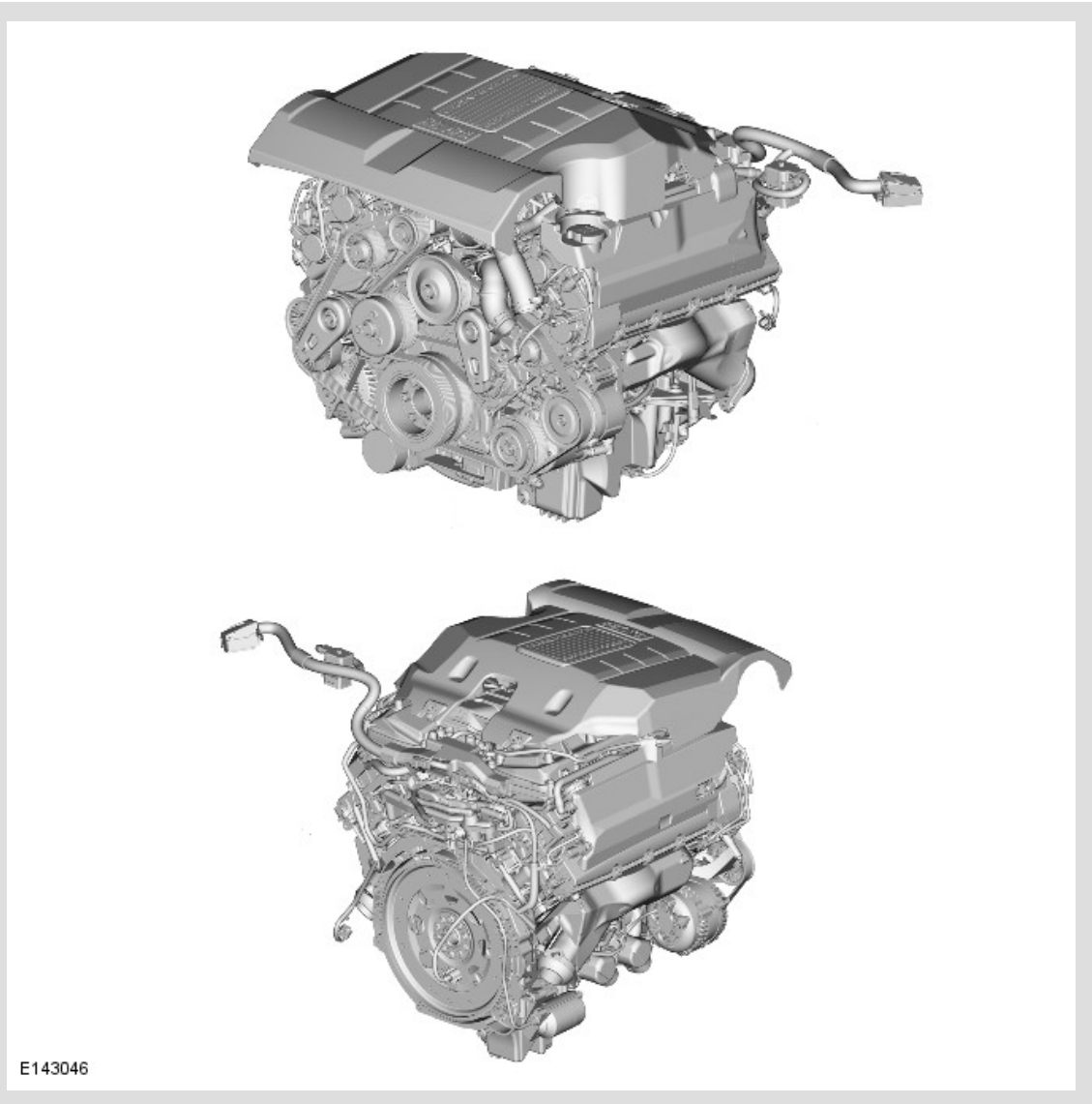
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2014.0 RANGE ROVER (LG), 303-01

ENGINE - V8 S/C 5.0L PETROL

DESCRIPTION AND OPERATION

EXTERNAL VIEWS

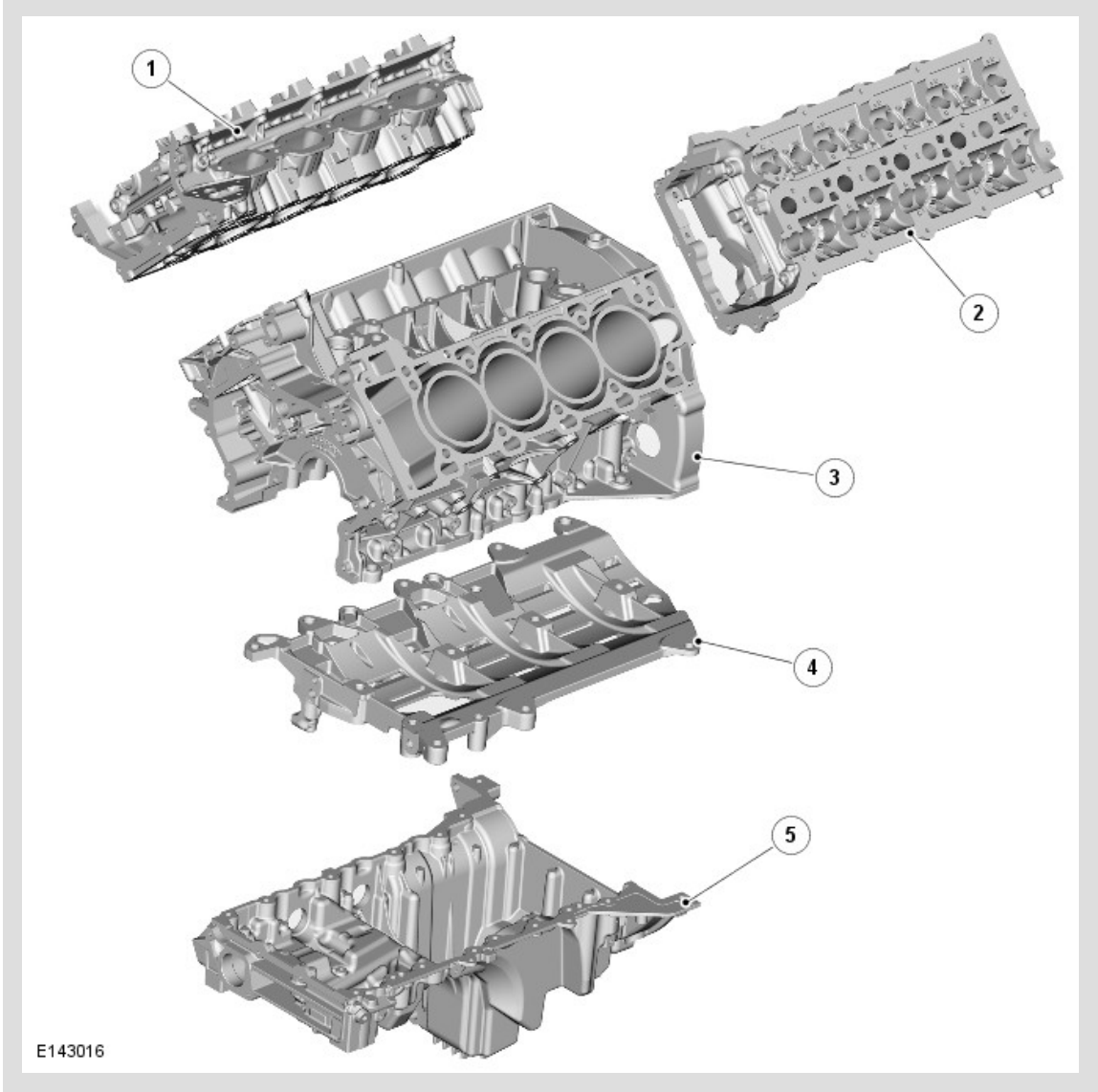


OVERVIEW

The 5.0L SC (supercharger) gasoline engine is a liquid cooled V8 unit featuring direct fuel injection, four overhead camshafts and four valves per cylinder. All four camshafts incorporate VCT (variable camshaft timing) .

DESCRIPTION

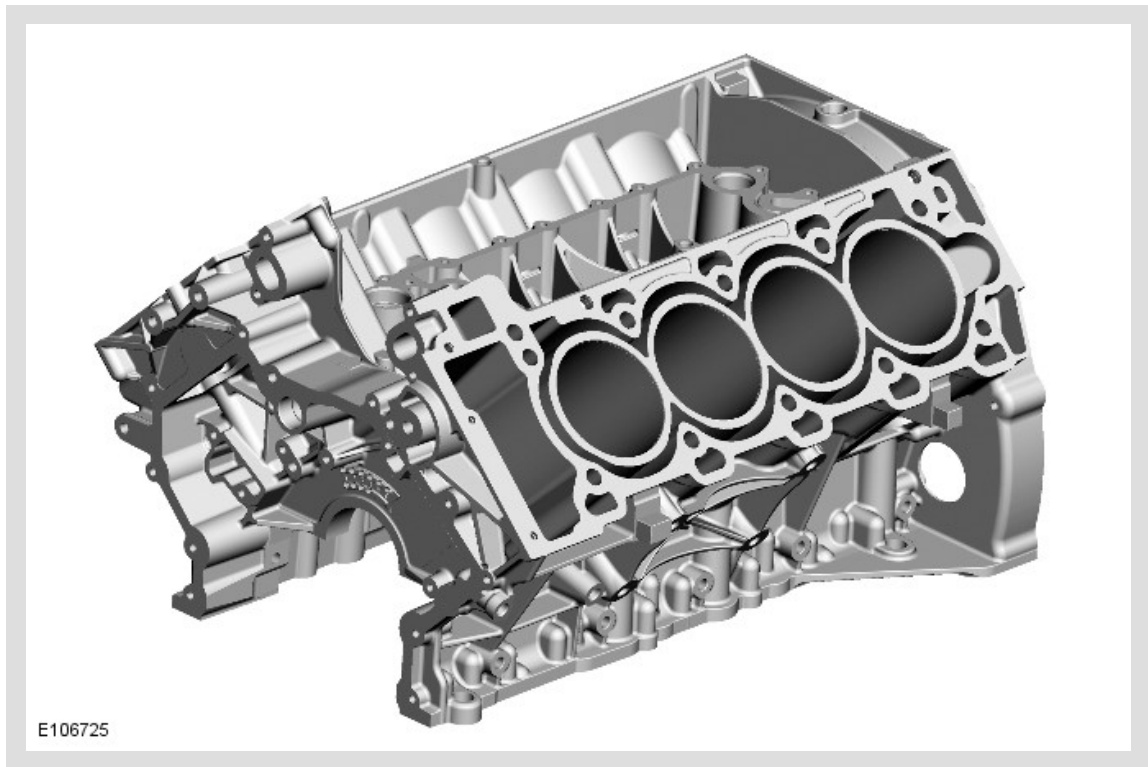
ENGINE STRUCTURE



ITEM	DESCRIPTION
1	Right cylinder head (bank A)
2	Left cylinder head (bank B)
3	Cylinder block
4	Windage tray
5	Oil pan

The engine is built around a very stiff, lightweight, enclosed V, deep skirt cylinder block. A structural windage tray is bolted to the bottom of the cylinder block to further improve the block stiffness, minimize NVH (noise, vibration and harshness) and help reduce oil foaming. To further enhance the stiffness of the lower engine structure, a heavily ribbed oil pan is installed. The oil pan also helps to reduce engine noise.

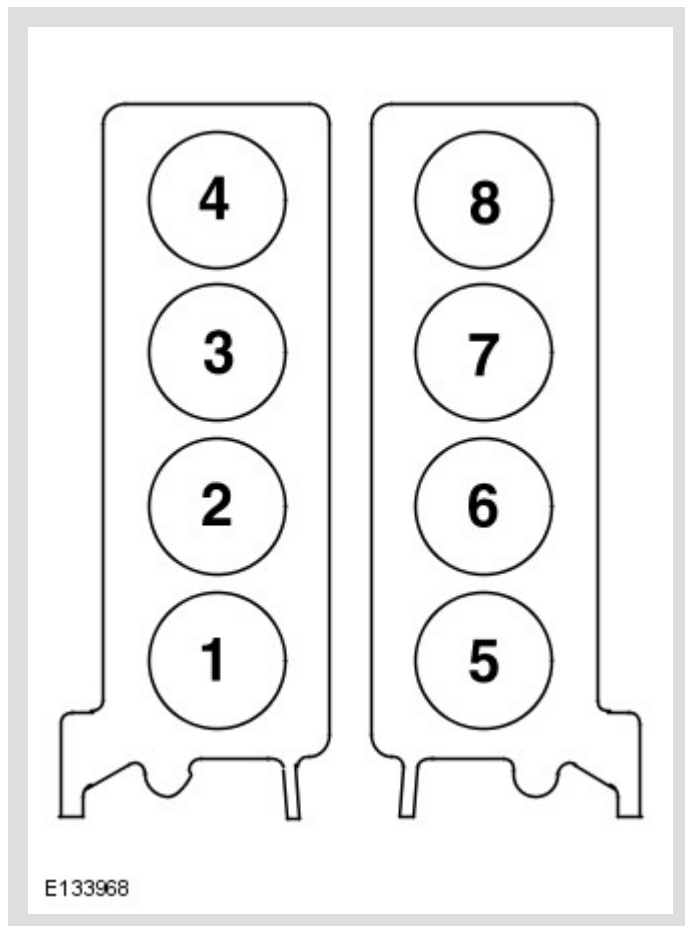
CYLINDER BLOCK



The cylinder block is a 90 degree configuration with cylinder liners and an open deck die-cast coolant jacket. The low volume coolant jacket gives good warm-up times and low piston noise levels. The longitudinal flow design of the coolant jacket, with a single cylinder head coolant transfer port in each bank, provides good rigidity and head gasket sealing.

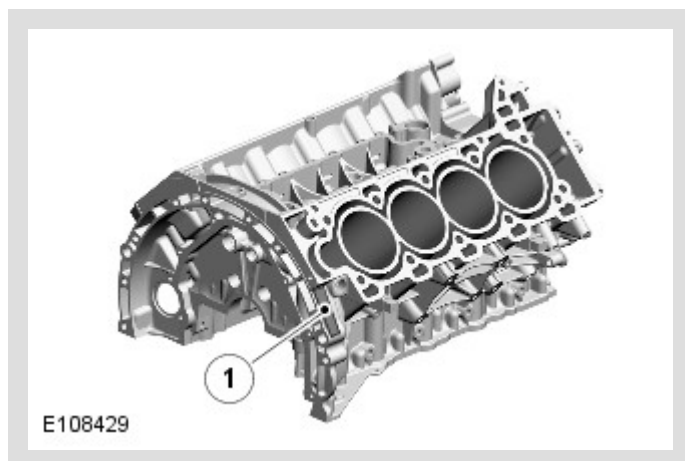
Cylinder Numbering

DIN standard cylinder numbering



DIN cylinder firing order 1,5,4,2,6,3,7,8

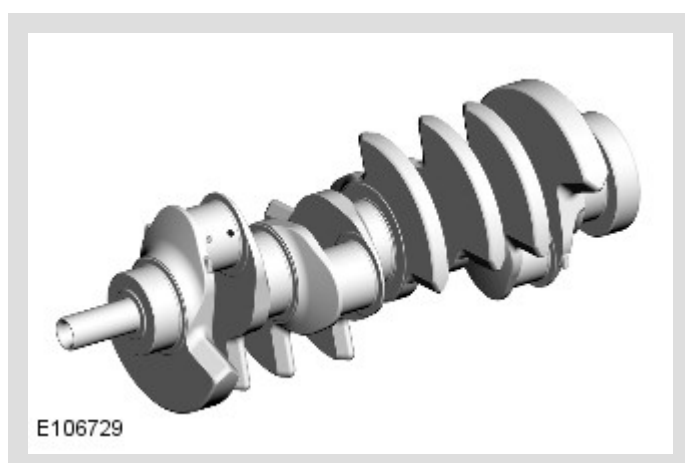
ENGINE DATA LOCATION



ITEM	DESCRIPTION
1	Engine data location

Engine data is marked on the cylinder block at the rear of the right cylinder bank.

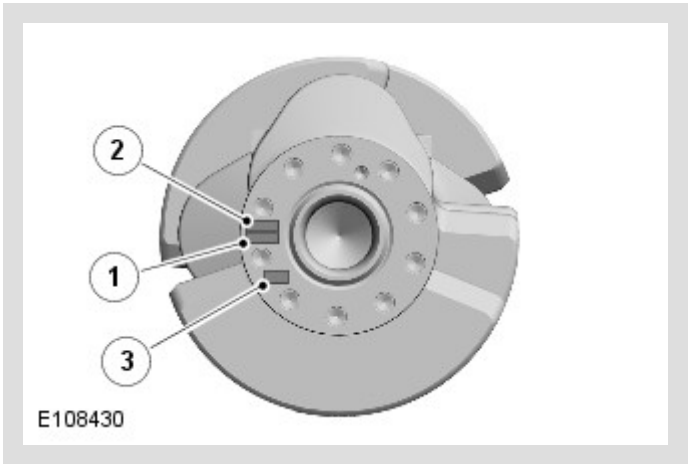
CRANKSHAFT



The crankshaft is installed in five main bearings. Eight counter-balance weights ensure low vibration levels and the large main bearing journals are designed to contribute to stiffness.

An oil groove in the upper half of each main bearing transfers the oil into the crankshaft for lubrication of the connecting rod bearings. A thrust washer is installed each side of the top half of the center main bearing.

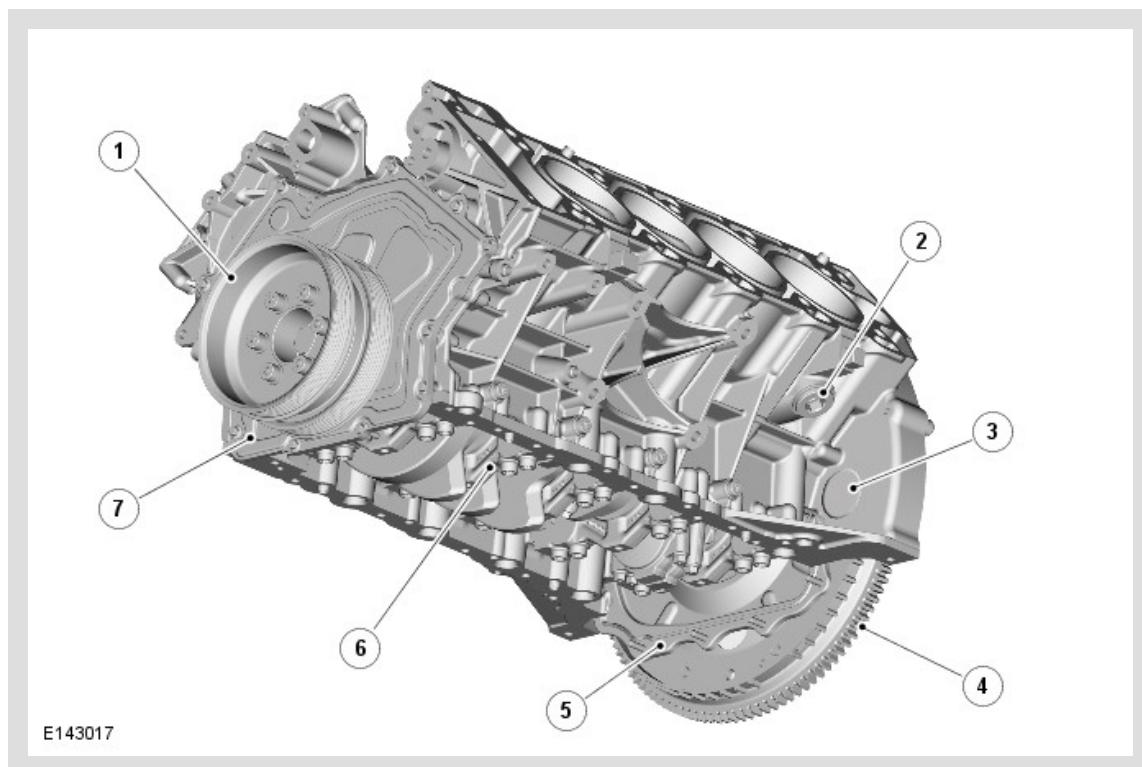
CRANKSHAFT DATA LOCATION



ITEM	DESCRIPTION
1	Pin journal classification and plant identification
2	Main journal classification
3	Date and time codes

The main bearings are numbered 1 to 5 starting from the front of the engine. There are five grades of main bearing available, each being color coded. Journal sizes are marked on the rear of the crankshaft.

CRANKSHAFT INSTALLATION



ITEM	DESCRIPTION
1	Front pulley
2	Coolant drain plug
3	Torque converter access plug
4	Drive plate
5	Rear cover
6	Main bearing cap
7	Front cover

The main bearing caps are cross bolted to increase rigidity. An identification mark on the bearing cap faces the front of the engine.

At the front of the crankshaft, a tuned torsional vibration damper is incorporated into the crankshaft front pulley. At the rear of the crankshaft a drive plate, with a starter ring gear, is installed to transfer drive from the engine to the transmission. The

reluctor ring for the CKP (crankshaft position) sensor is attached to the front of the drive plate.

The crankshaft seals are located in the front and rear covers.

PISTONS AND CONNECTING RODS

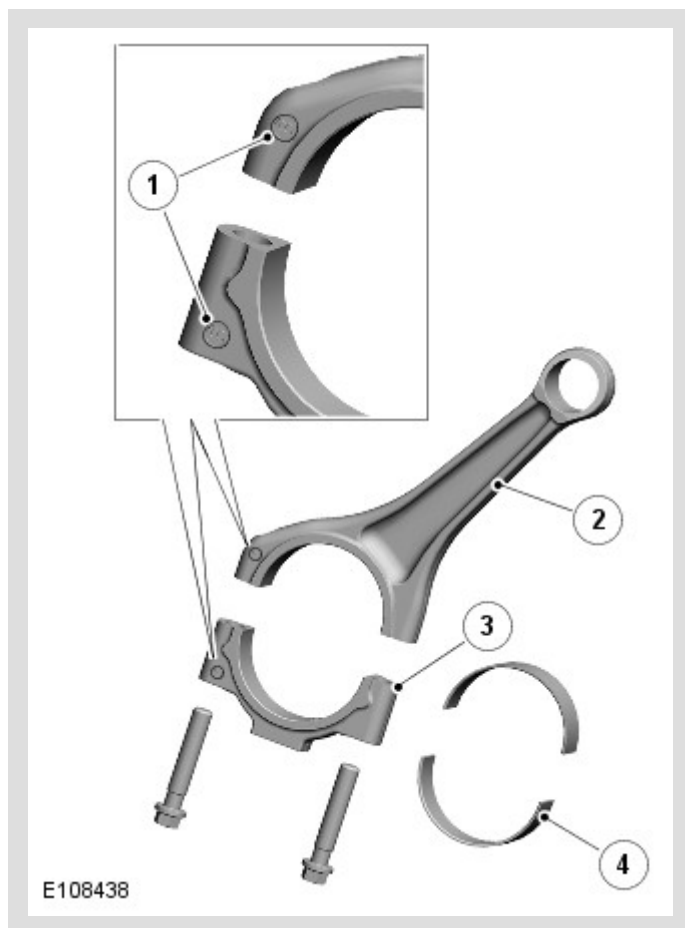


The diameter of each piston is graded and precisely matched to each cylinder bore to help reduce noise. In the vertical plane, the pistons have a slight barrel form, which helps to ensure a reliable oil film is maintained between the piston and the cylinder bore. A solid film lubricant coating is applied to both reaction faces of the piston to reduce wear and improve fuel economy.

A three-ring piston-sealing system is used. The top ring is treated with a PVD (physical vapor deposition) peripheral coating. PVD is a coating technique where material can be deposited with improved properties to ensure good cylinder bore compatibility and wear resistance. A Napier center ring helps cylinder pressure and oil management. The lower ring is a three-piece oil control.

The connecting rod caps are fracture-split from the rods to ensure precision re-assembly for bearing shell alignment. There are three grades of large end bearing available, each being color coded.

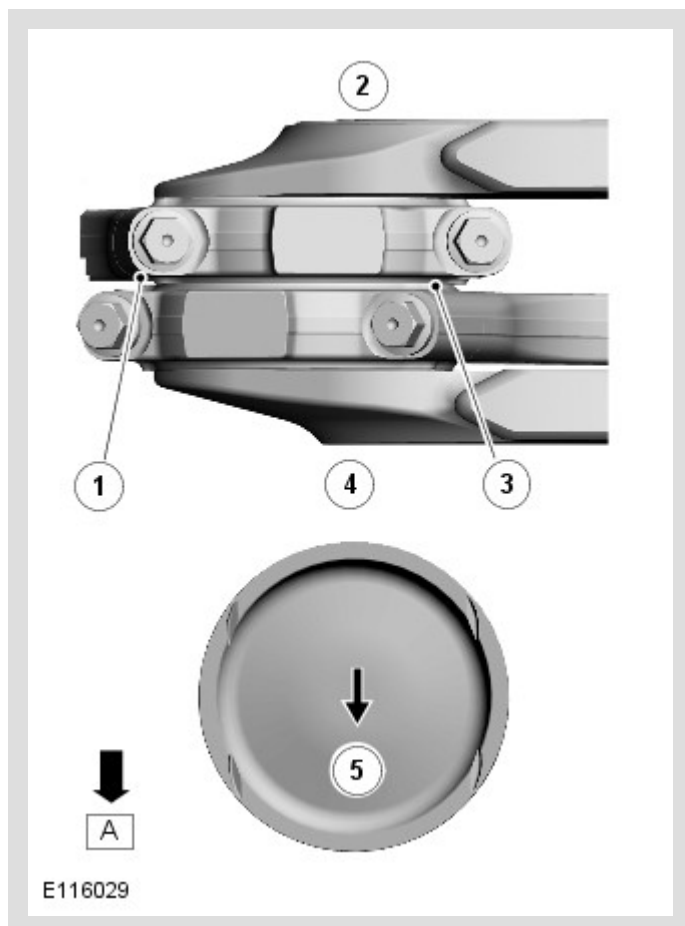
CAP ALIGNMENT WITH CONNECTING ROD



ITEM	DESCRIPTION
1	Alignment marks
2	Connecting rod
3	Cap
4	Bearings

The correct alignment of the cap with the connecting rod is indicated by marks on adjacent faces of the two components.

CONNECTING ROD AND PISTON ORIENTATION



ITEM	DESCRIPTION
A	Front of engine
1	Alignment mark
2	Left side (bank B)
3	Alignment mark
4	Right side (bank A)
5	Piston orientation arrow

The orientation of the connecting rods and pistons on the crankshaft are given below:

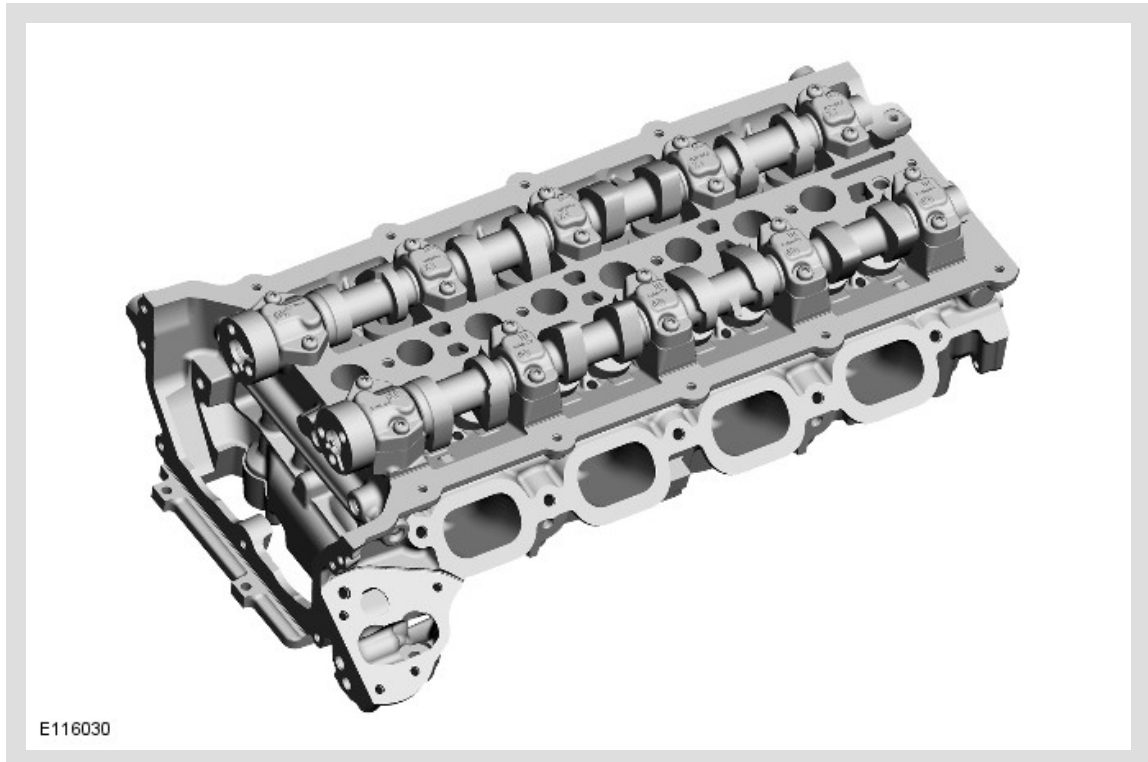
- Bank A - The arrow on the piston crown must face the front of the engine and the cap and connecting rod alignment marks must face the rear of the engine.

- Bank B - The arrow on the piston crown must face the front of the engine and the cap and connecting rod alignment marks must face the front of the engine.

CYLINDER HEADS

NOTE:

Right (A bank) cylinder head shown, left (B bank) cylinder head similar.



The cylinder heads are unique for each cylinder bank. Deep-seated bolts reduce distortion and secure the cylinder heads to the cylinder block.

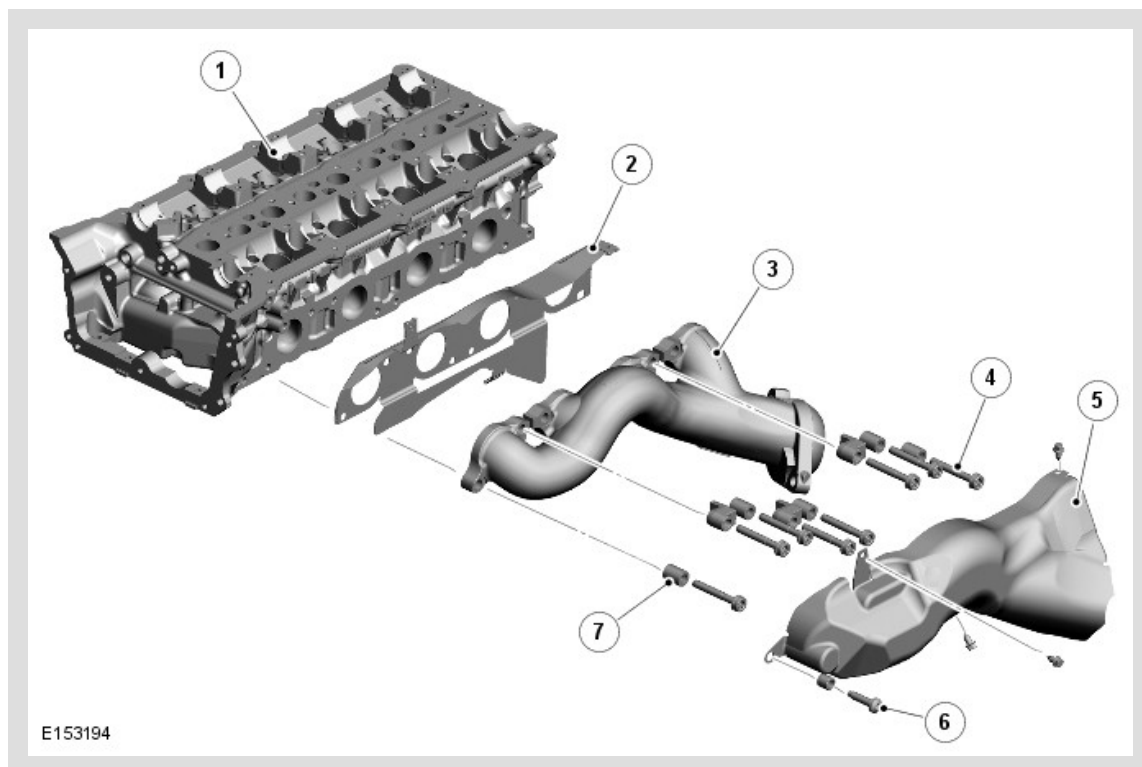
Each cylinder is served by four valves. To help achieve the required gas-flow characteristics, these are arranged asymmetrically around the cylinder bore. Each cylinder has a centrally mounted fuel injector and spark plug.

The cylinder head gasket is of a multi-layer construction.

EXHAUST MANIFOLD

NOTE:

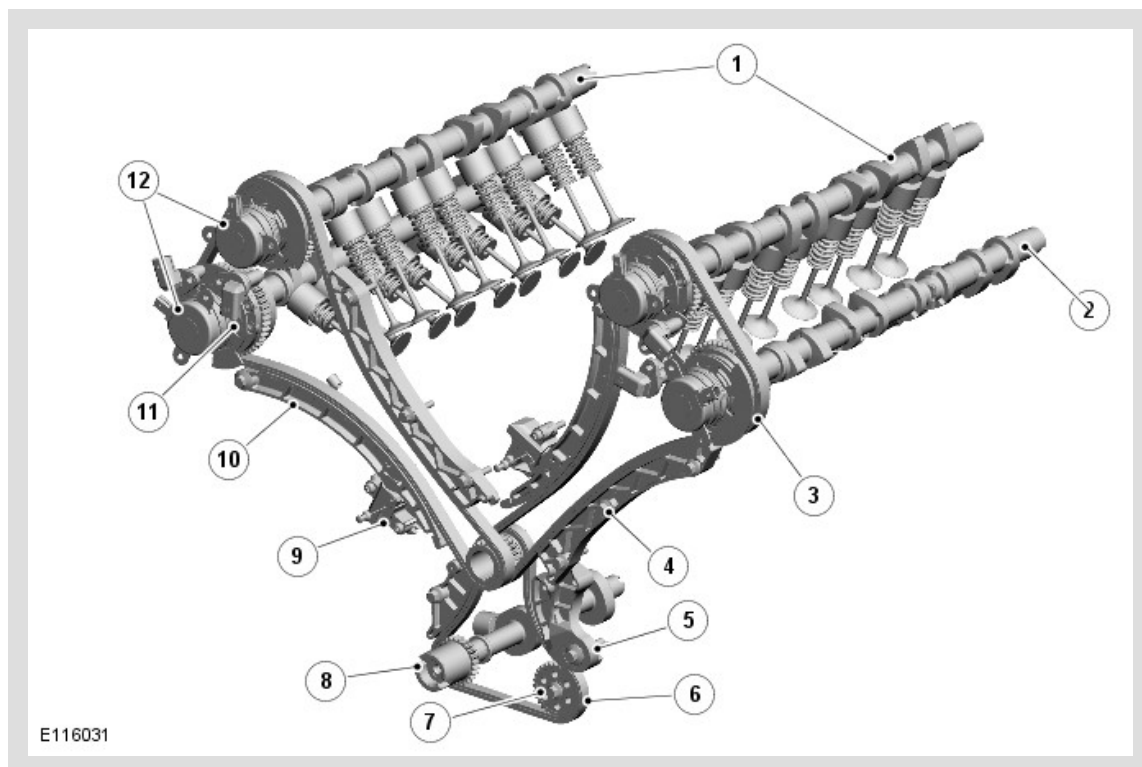
Left (B bank) installation shown, right (A bank) installation similar.



ITEM	DESCRIPTION
1	Cylinder head
2	Gasket
3	Exhaust manifold
4	Bolt (8 off)
5	Heat shield
6	Bolt (4 off)
7	Spacer (8 off)

The manifolds are unique for each cylinder bank. Each exhaust manifold installation includes two gaskets and two heat shields. Spacers on the securing bolts allow the manifolds to expand and contract with changes of temperature while maintaining the clamping loads.

VALVE TRAIN



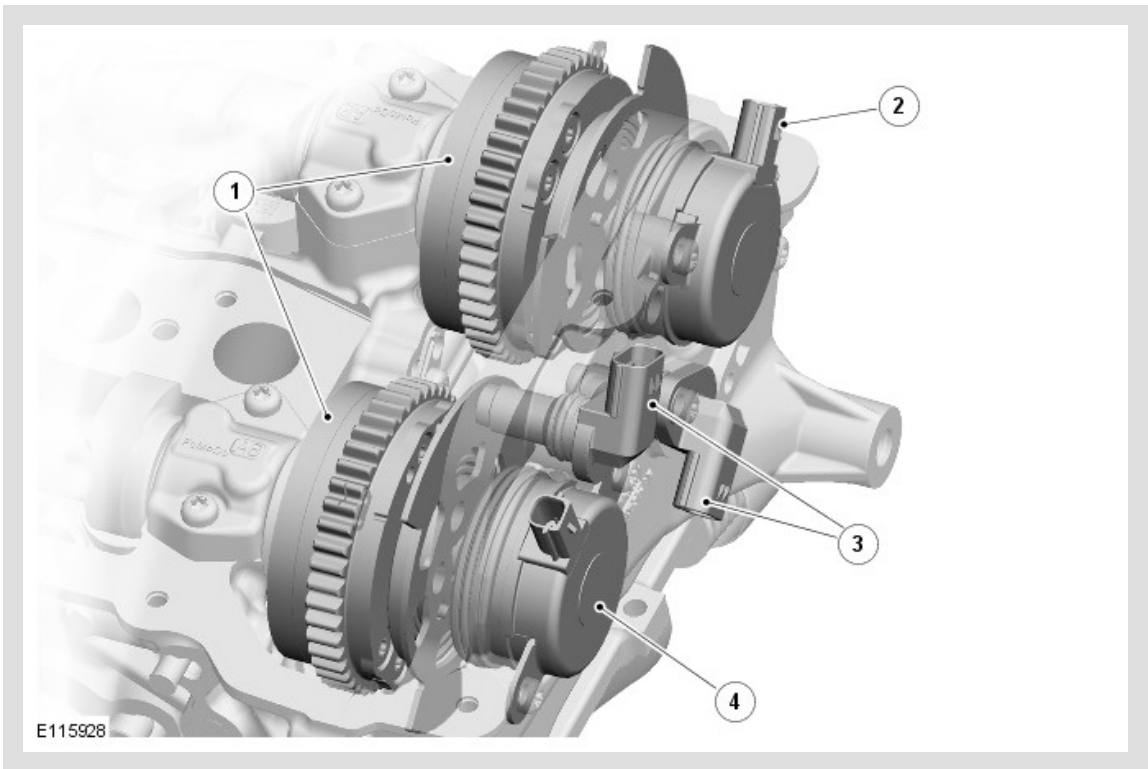
ITEM	DESCRIPTION
1	Intake camshafts
2	Exhaust camshaft
3	Inverted tooth timing chain
4	Chain guide
5	Auxiliary chain tensioner
6	Auxiliary drive chain
7	Oil pump drive
8	Auxiliary drive camshaft
9	Timing chain tensioner
10	Tensioner lever
11	Variable camshaft timing actuator
12	Variable camshaft timing solenoids

The lightweight valve train provides good economy and noise levels and is chain driven from the crankshaft.

Double overhead camshafts on each cylinder head operate the valves. For each cylinder head, an inverted tooth timing chain transfers drive from the crankshaft to the VCT actuator on the front of each camshaft. Graded tappets enable setting of inlet and exhaust valve clearances.

Each timing chain has a hydraulic tensioner operated by engine oil pressure. The chain tensioners incorporate a ratchet mechanism, which maintains tension while the engine is stopped to eliminate start-up noise. The chains are lubricated with engine oil from jets located at the front of the engine block. Chain guides control chain motion on the drive side.

VARIABLE CAMSHAFT TIMING



ITEM	DESCRIPTION
1	Variable camshaft timing actuators
2	Intake variable camshaft timing solenoid
3	Camshaft position sensors
4	Exhaust variable camshaft timing solenoid

The VCT system varies the timing of the intake and exhaust camshafts to deliver optimum engine power, efficiency and emissions. The timing of the intake camshafts

has a range of 62 degrees of crankshaft angle. The timing of the exhaust camshafts has a range of 50 degrees of crankshaft angle.

In the base timing position:

- The intake camshafts are fully retarded.
- The exhaust camshafts are fully advanced.

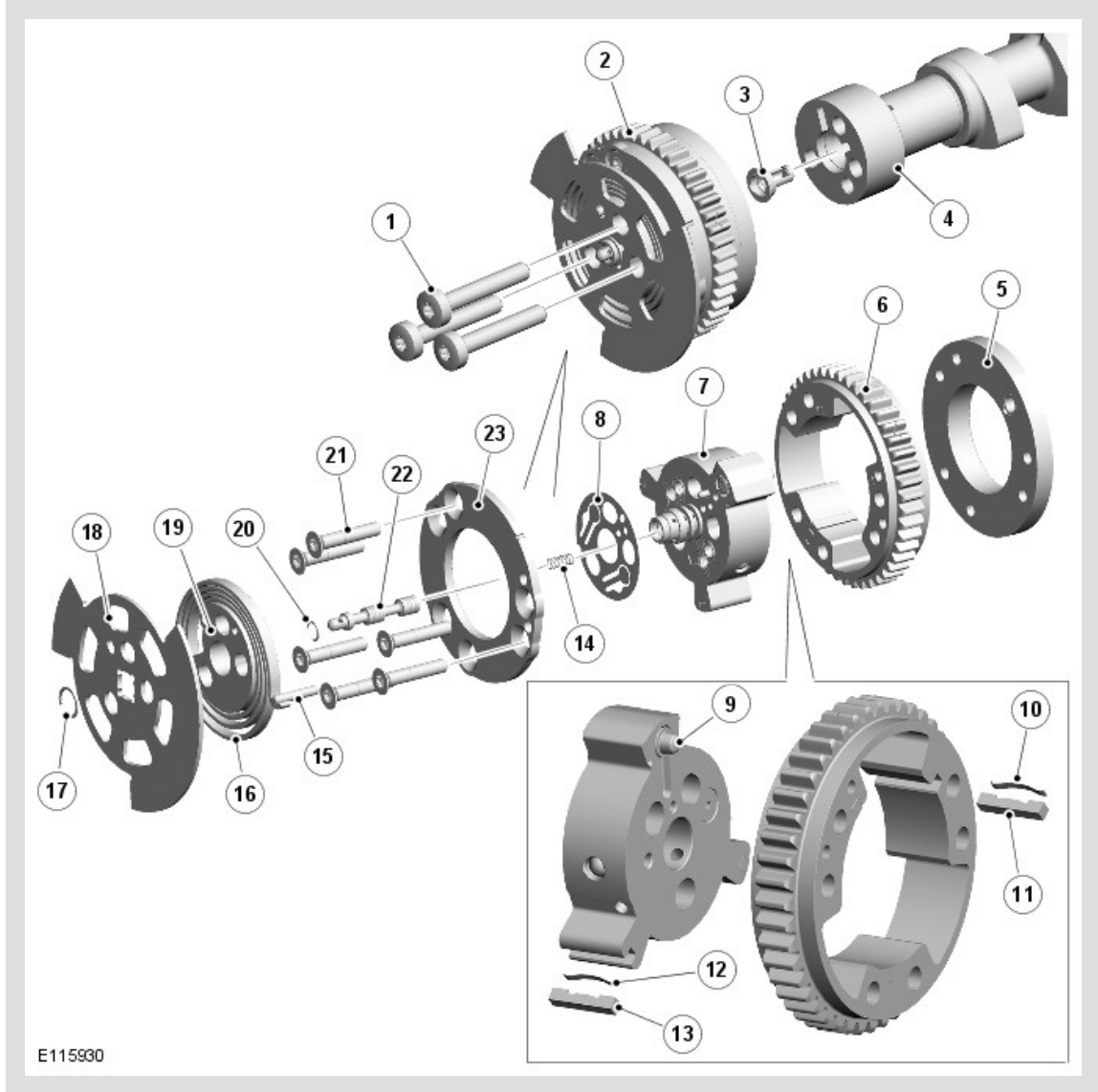
VCT Operating Ranges

CAMSHAFT	VALVE OPENS	VALVE CLOSES
Intake	29 degrees BTDC (before top dead center) to 33 degrees ATDC (after top dead center)	207 to 269 degrees ATDC
Exhaust	244 to 194 degrees BTDC	6 to 56 degrees ATDC

The system consists of a VCT actuator and a VCT solenoid for each camshaft. The ECM (engine control module) controls the system using PWM (pulse width modulation) signals to the VCT solenoids.

The torsional energy generated by the valve springs and the inertia of the valve train components are used to operate the system.

VARIABLE CAMSHAFT TIMING ACTUATORS



ITEM	DESCRIPTION
1	Bolt (3 off)
2	Variable camshaft timing actuator
3	Filter
4	Camshaft
5	Inner plate
6	Housing and sprocket
7	Rotor assembly

8	Reed plate
9	Spring and lock pin
10	Spring (3 off)
11	Tip seal (3 off)
12	Spring (2 off)
13	Tip seal (2 off)
14	Spring
15	Dowel pin
16	Bias spring
17	Snap ring
18	Reluctor ring
19	Center plate
20	Snap ring
21	Screw (6 off)
22	Spool valve
23	Outer plate

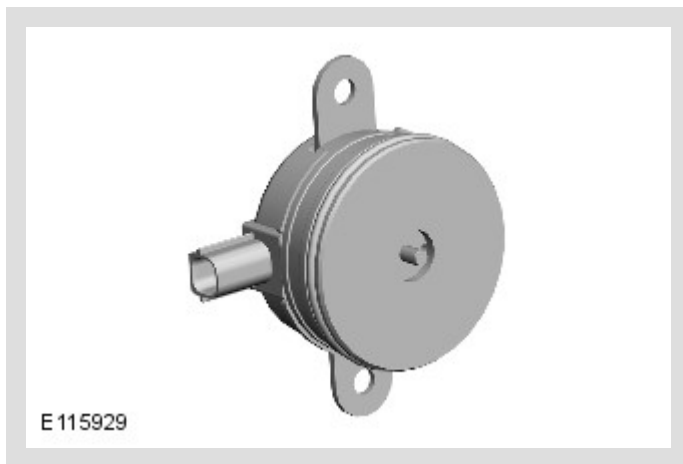
The VCT actuators change the position of the camshafts in relation to the timing chains.

Each VCT actuator is attached to the camshaft by three bolts. A rotor assembly and a reed plate are installed inside a sprocket housing, which consists of a sprocket, an outer plate and an inner plate held together by six screws.

A reluctor ring, for the CMP (camshaft position) sensor, a center plate and a bias spring are installed at the front of the VCT actuator. The ends of the bias spring locate on the center plate assembly and the sprocket housing, to give a turning moment to the camshaft in the advance direction. A snap ring locates the reluctor ring on to a sleeve installed in the center of the rotor assembly. The opposite end of the sleeve locates in a bore in the front face of the camshaft, which contains a filter.

A spring and spool valve are installed in the rotor assembly sleeve and retained by a snap ring. The spring keeps the spool valve in contact with the armature of the related VCT solenoid.

Each VCT actuator is supplied with engine oil from an oil gallery in the cylinder head, through the camshaft front bearing cap and a bore in the center of the camshaft.



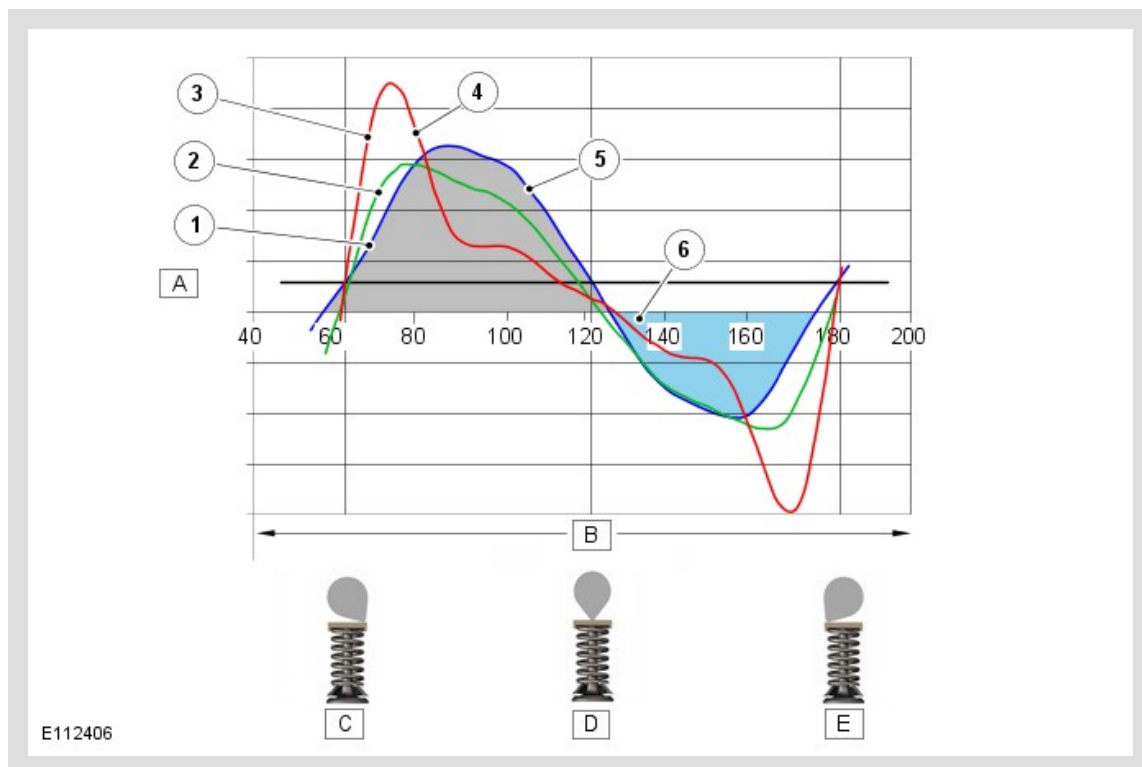
The VCT solenoids control the position of the spool valves in the VCT actuators.

The VCT solenoids are installed in the front upper timing covers, immediately in front of their related VCT actuators. Each VCT solenoid is secured with two screws and sealed with an O-ring. A two pin electrical connector provides the interface with the engine harness.

Each VCT solenoid incorporates a spindle that acts on the spool valve in the related VCT actuator to advance and retard the camshaft timing. The VCT solenoids operate independently and are controlled by a PWM signal from the ECM .

When the engine is running, the compression and expansion of the valve springs causes momentary increases and decreases in the torque acting on the camshafts. These momentary changes of torque are sensed in the VCT actuators and used to change the camshaft timing.

Camshaft Torsional Energy (For a Single Valve Event)

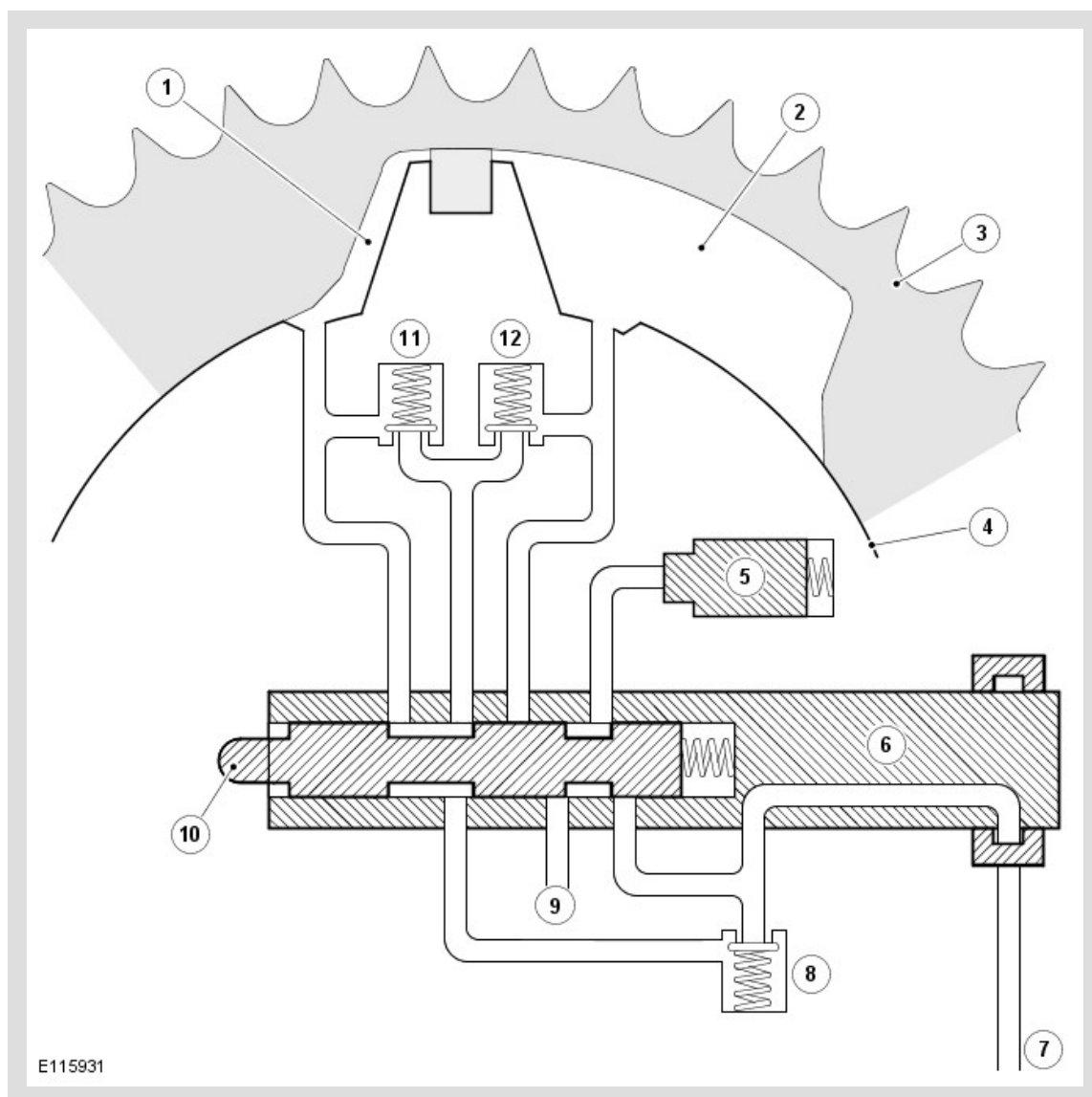


ITEM	DESCRIPTION
A	Camshaft torque
B	Camshaft rotation (degrees)
C	Valve opening
D	Peak lift
E	Valve closing
1	1000 rev/min
2	4000 rev/min
3	7000 rev/min
4	Inertia effects from valve train rotating components
5	Force caused by valve spring
6	Bias torque from friction

Variable Camshaft Timing Actuator Schematic - Base Timing

NOTE:

Intake camshaft VCT actuator shown. For exhaust camshaft VCT actuator, read advance for retard and retard for advance.



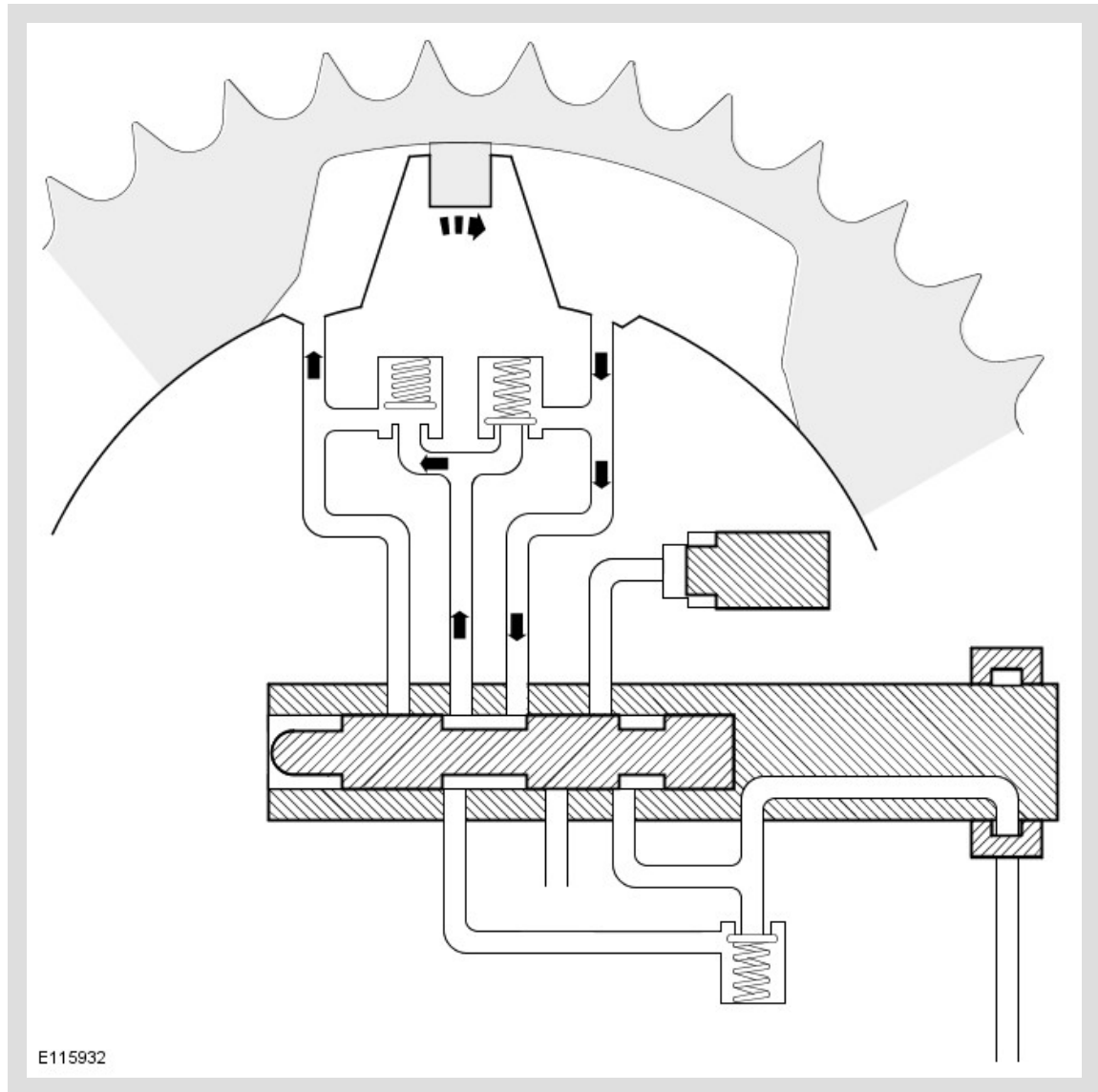
ITEM	DESCRIPTION
1	Advance chamber
2	Retard chamber
3	Sprocket housing
4	Rotor assembly
5	Lock pin
6	Sleeve
7	Engine oil supply from camshaft

8	Inlet check valve
9	Lock pin drain
10	Spool valve
11	Advance check valve
12	Retard check valve

At engine start-up, once the engine oil pressure in the camshaft is sufficient to open the inlet check valve, engine oil flows across the spool valve, through the advance and retard check valves and into the advance and retard chambers. During the start cycle, the ECM signals the VCT solenoid to move the spool valve into the sleeve and connect the lock pin to inlet oil pressure. The inlet oil pressure causes the lock pin to retract from the inner plate and unlock the rotor assembly and camshaft from the sprocket housing.

There is a constant supply of oil to the VCT to ensure the actuator remains filled during operation.

Variable Camshaft Timing Actuator Schematic - Advance

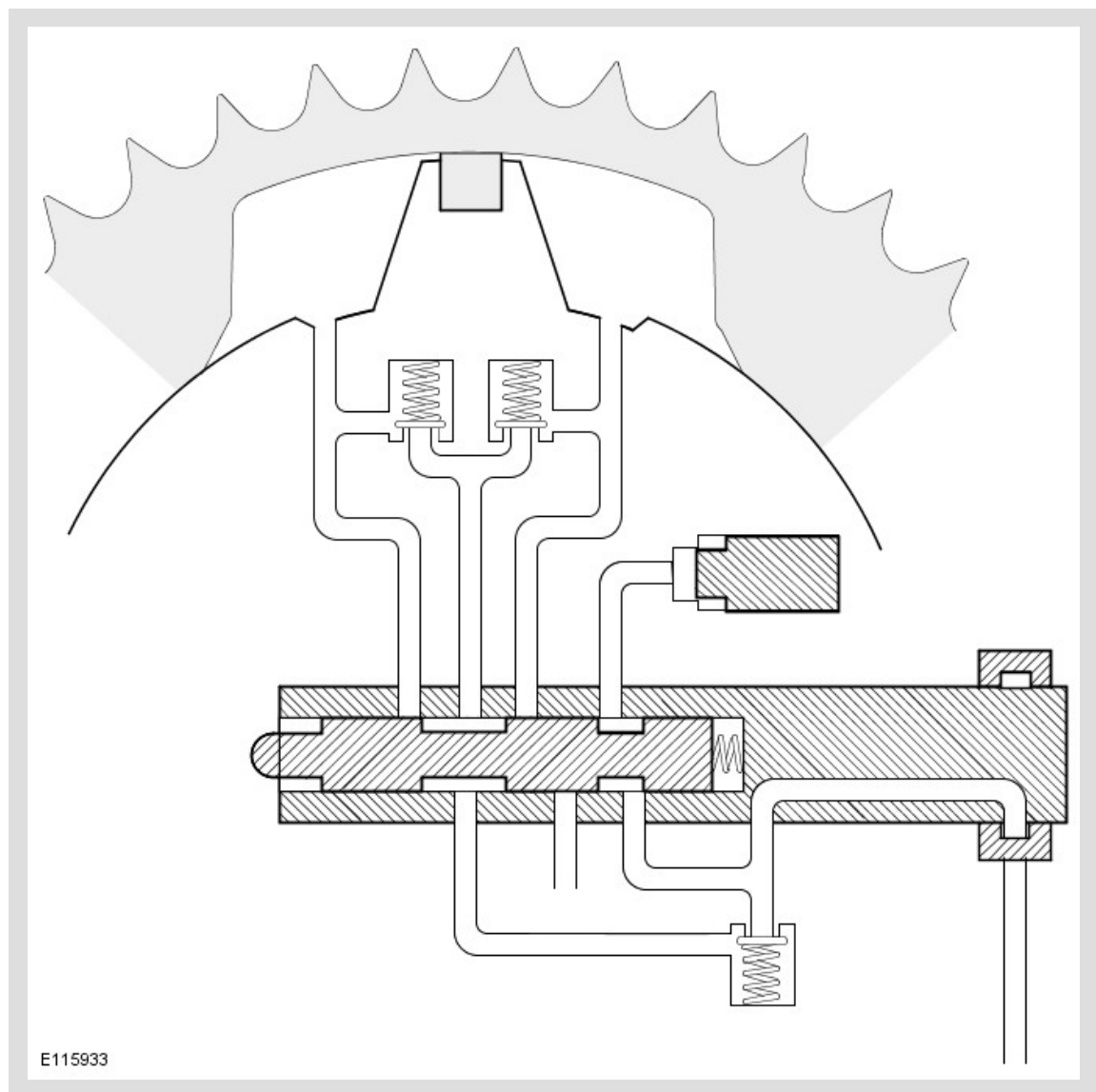


To advance the camshaft timing, the ECM adjusts the signal to the VCT solenoid to move the spool valve so that the advance chamber oil passage is closed and the retard chamber oil passage is connected to inlet oil.

Each momentary increase of the torque acting on the camshaft generates a pressure pulse in the retard chamber. Oil moves from the retard chamber, through the spool valve and the advance check valve to the advance chamber, to equalize the pressures in the two chambers. The displacement of oil from the retard chamber causes the rotor assembly to advance in relation to the sprocket housing. Each momentary decrease of torque acting on the camshaft also generates a pressure pulse in the advance chamber, but, with the advance chamber oil passage closed, no movement of oil

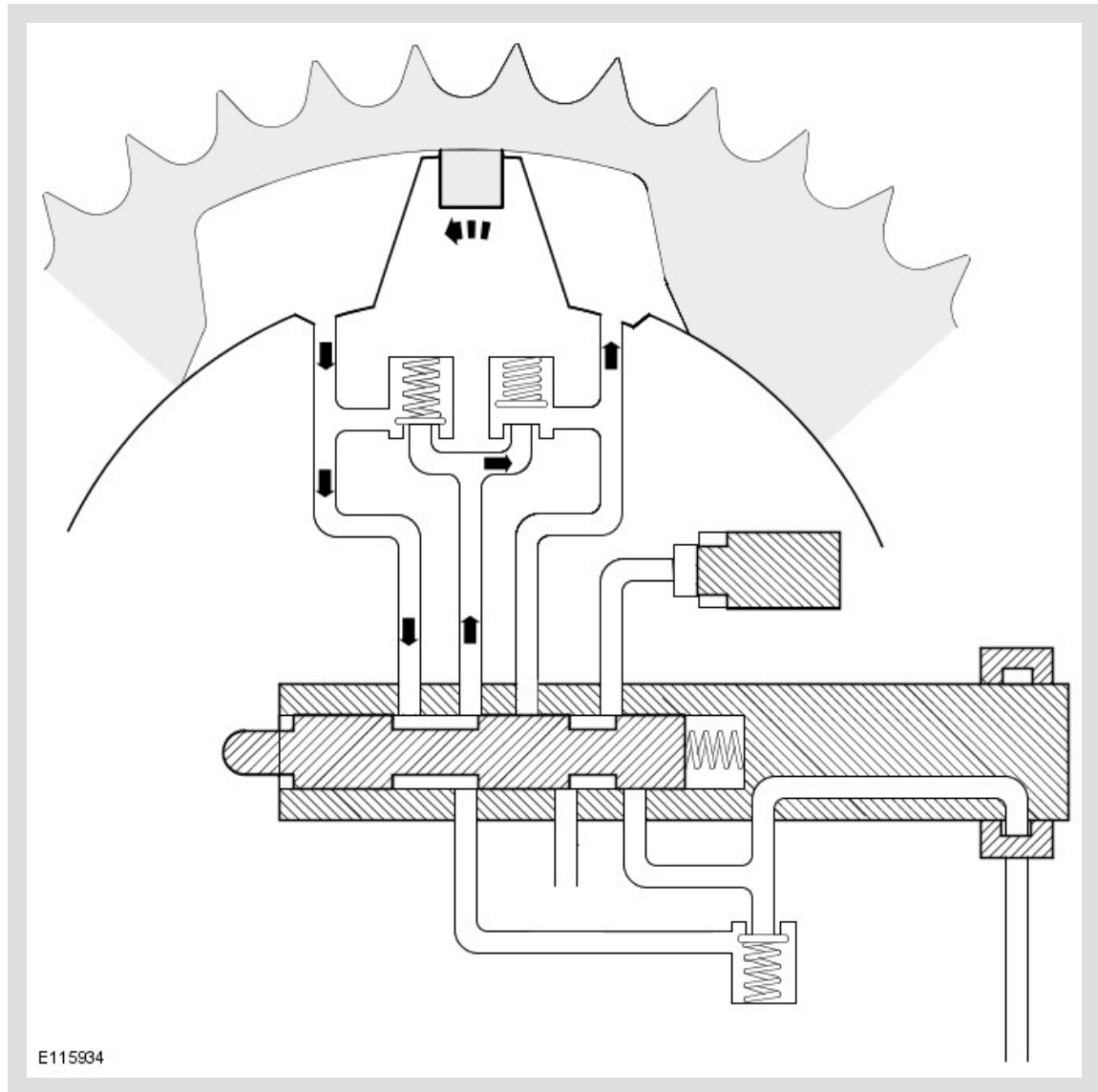
between the advance and retard chambers occurs and the rotor assembly cannot move in the retard direction.

Variable Camshaft Timing Actuator Schematic - Null



Once the camshaft has reached the required timing position the ECM adjusts the signal to the VCT solenoid to set the spool valve in the null position. In the null position, the advance and retard chamber oil passages are both closed by the spool valve and the rotor assembly is hydraulically locked to the sprocket housing.

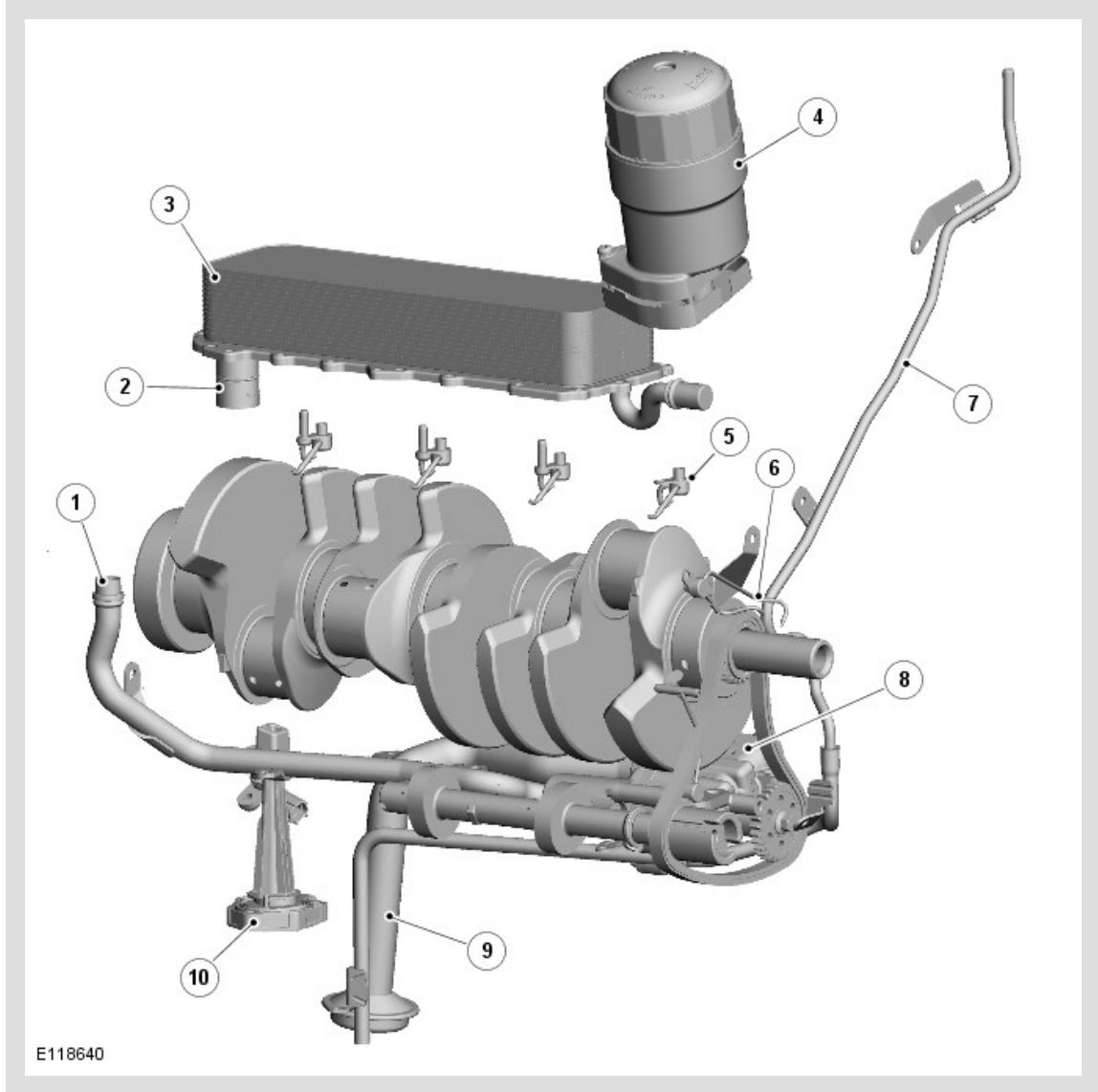
Variable Camshaft Timing Actuator Schematic - Retard



To retard the camshaft timing, the ECM adjusts the signal to the VCT solenoid to move the spool valve to close the retard chamber oil passage and connect the advance chamber oil passage to the inlet oil.

Each momentary decrease of the torque acting on the camshaft causes oil to transfer from the advance chamber, through the spool valve and the retard check valve to the retard chamber, and so retard the camshaft timing.

LUBRICATION SYSTEM



ITEM	DESCRIPTION
1	Oil pump outlet tube
2	Anti-drain valve
3	Oil cooler
4	Oil filter
5	Piston cooling jet (4 off)
6	Timing chain lubrication jet (2 off)
7	Oil evacuation tube

8	Oil pump
9	Oil pick-up
10	Oil level and temperature sensor

The oil pump is attached to the underside of the windage tray. The input shaft of the oil pump is driven from the front of the crankshaft, by the auxiliary chain, at 0.87 engine speed.

The oil pump draws oil from the sump through a centrally mounted pick-up pipe. The oil is pressurized and pumped through an output tube to the cylinder block. After passing through an anti-drain valve and a plate type oil cooler, the oil is filtered by a replaceable cartridge installed on the front of the RH (right-hand) cylinder head.

The output from the oil filter is distributed through oil galleries in the cylinder heads and cylinder block. All moving parts are lubricated by pressure or splash oil. Pressurized oil is also provided for the VCT system, the timing chain tensioners, the piston cooling jets and the timing chain lubrication jets.

The oil returns to the oil pan under gravity. Large drain holes through the cylinder heads and cylinder block ensure the rapid return of the oil to the sump. System replenishment is through the oil filler cap on the left cylinder head cover.

An oil evacuation tube is installed to allow oil to be drawn from the sump. The upper end of the oil evacuation tube is located under the oil filler cap.

An oil drain plug is installed in the right side of the sump.

Oil Pump Nominal Operating Pressures

ENGINE SPEED, REV/MIN	TEMPERATURE, °C (°F)	PRESSURE, BAR (LBF/IN ²)
Idle	20 (68)	2.0 (29.0)
1500	20 (68)	6.0 (87.0)
3000	40 (104)	6.2 (90.0)
3000	110 (230)	5.0 (72.5)
3000	130 (266)	4.0 (58.0)

OIL LEVEL MONITORING

Oil level monitoring is provided by an oil level and temperature sensor that measures the oil level in the sump. The oil level can be displayed in the message center of the instrument cluster.

Oil Level and Temperature Sensor



The oil level and temperature sensor supplies the ECM with a signal containing the level and temperature of the oil in the sump. The oil level and temperature sensor is secured to the bottom of the sump with three screws and sealed with an O-ring.

The oil level and temperature sensor sends an ultrasonic pulse vertically upward and measures the time taken for the pulse to be reflected back from the top surface of the oil. This time is compared with the time taken for an ultrasonic pulse to travel a reference distance within the oil level and temperature sensor to determine the oil level. The oil level reading is combined with the oil temperature reading and transmitted in a PWM signal to the ECM .

Oil Level and Temperature Sensor Specifications

FEATURE	DETAILS

Power Source	Battery Voltage
Level Accuracy	±2 mm (±0.08 in.) at temperatures of 30 °C (86 °F) and above
	±4 mm (±0.16 in.) at temperatures between -10 and +30 °C (14 and 86 °F)
Temperature Accuracy	±2 °C (±3.6 °F)
Operating Level Range	18 to 147 mm (0.71 to 5.79 in.)
Operating Temperature Range	-40 to 160 °C (-40 to 320 °F)

OIL LEVEL CHECK

For accuracy, oil level checks should be performed with the vehicle on level ground when the oil is hot. The vehicle needs to stand for approximately 10 minutes, after the engine is switched off, to allow the oil to drain back into the sump and the oil level to stabilize. The oil level system will not give a reading until the oil level has stabilized.

The method of checking the oil level can be found in the oil draining and filling procedures. For additional information, refer to: (303-01 Engine - V8 S/C 5.0L Petrol)

Engine Oil Draining and Filling (General Procedures),
Engine Oil Vacuum Draining and Filling (General Procedures).

OPERATION

Operation of the engine is controlled by the ECM .